

Jotheesh Reddy Kummathi

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EXPERIENCE

AI Vision Systems Intern

July 2026 – Present

The Nail Lab, Inc.

New York, NY

- Developing embedded vision firmware on NVIDIA Jetson Orin Nano (ARM64 Linux) in C++ and Python, running real-time detection, segmentation, and tracking on-device.
- Built a GStreamer video pipeline ingesting dual IMX219 sensors over MIPI CSI-2 with hardware-accelerated encode; synchronized capture holds frame alignment across overlapping views.

Graduate Research Assistant

Jan. 2025 – June 2026

Agile Robotics and Perception Lab (ARPL) – New York University

New York, NY

- Deployed and calibrated a multi-modal sensor suite (LiDAR, depth cameras, precision GPS) for state estimation in GPS-denied environments; contributed to 3D Gaussian Splatting perception research.
- Brought up PX4 stacks on Jetson Orin companions, root-caused ESC and flight-controller faults reproducible only on hardware, and built HITL/SITL CI/CD pipelines with telemetry logging.

Head Teaching Assistant

Jan. 2025 – May 2026

New York University – ECE-GY 6483: Real-Time Embedded Systems

New York, NY

- Led recitations for 700+ graduate students over three semesters on C, Embedded C, Mbed OS, RTOS scheduling, and real-time DSP; graded labs on ARM Cortex-M peripherals and interrupts.

Junior Embedded Engineer

July 2023 – July 2024

Magnibot Technology Solutions Pvt. Ltd.

Bengaluru, India

- Developed and optimized firmware for IoT and embedded products in C/C++ on FreeRTOS, shipped into domestic and industrial deployments.
- Designed, debugged, and integrated embedded solutions across UART, SPI, I2C, and CAN; maintained CI/CD pipelines automating build validation and regression testing.

PROJECTS

Real-Time Tremor & Dyskinesia Detection on STM32 | Mbed OS, CMSIS-DSP, BLE

Jan. 2026

- Wrote Mbed OS firmware on an STM32 B-L475E-IOT01A (Cortex-M4) that samples the onboard LSM6DSL gyroscope continuously, filters the angular-rate stream, and runs a CMSIS-DSP FFT on-device – all spectral analysis stays on the MCU, with no host PC in the loop.
- Separated Parkinsonian tremor (3–5 Hz) from dyskinesia (5–7 Hz) by spectral band, graded severity from peak magnitude, and published results over BLE GATT notifications with on-board LED status for untethered wear.

Real-Time Control Loop on Boston Dynamics Spot | C++/Python, ZED 2i, YOLOv8

Jan. 2026

- Deployed closed-loop person-following to physical Boston Dynamics Spot hardware: YOLOv8 detections on a ZED 2i stereo camera drive a 3-axis proportional controller (range, lateral offset, yaw) at 10 Hz, with deadband filtering and velocity ramping to suppress command chatter near the setpoint.

Slip-Aware MPPI Navigation for Skid-Steer Robots | UKF, MPPI, ROS2, 3D LiDAR

Sep. – Dec. 2025

- Built the estimation and control stack for a Clearpath Warthog skid-steer: a UKF fusing wheel odometry, IMU, and 3D LiDAR for GPS-denied localization, feeding an MPPI controller sampling 1500 rollouts at 10 Hz.
- Measured a $2.2\times$ wheel-slip effect the nominal kinematic model missed (effective track width 1.38 m $\rightarrow$ 3.0 m); folding the measured slip into the plant model recovered path tracking on low-friction terrain.

TECHNICAL SKILLS

Languages & Build: C, C++, Embedded C, Python, Bash/Shell, GCC ARM, CMake, Git

MCUs & Firmware: STM32 (Cortex-M4), ESP32, Pixhawk, Bare-Metal, FreeRTOS, Mbed OS

Peripherals & Buses: UART, SPI, I2C, CAN/CAN-FD, BLE, MIPI CSI-2, MAVLink, DMA, Timers, ADC/PWM

Embedded Linux & DSP: Jetson Orin Nano (L4T ARM64), GStreamer, CMSIS-DSP, FFT, Sensor Fusion

Debug & Test: JTAG/SWD, GDB, STM32CubeMX, Oscilloscopes, Logic Analyzers, HITL/SITL Test Rigs

EDUCATION

New York University, New York, NY — M.S., Computer Engineering

Aug. 2024 – May 2026

Blekinge Institute of Technology, Sweden — B.S., Electrical Engineering (Exchange)

Aug. 2022 – May 2023

JNTUA College of Engineering, India — B.Tech., Electronics & Communication Engineering Aug. 2019 – Aug. 2022